

Homework 6 (Extra - not graded)

MATH 518 - Fall 2025

Unless stated otherwise, k is an algebraically closed field of characteristic zero.

1. What is $\dim(X)$ in the following cases?

1. $X = V(xy - 1) \subset k^2$,
2. $X = V(xy - z^2, x + w - 1) \subset k^4$
3. $X = \mathbb{P}^n(k)$
4. $X = V(xy - w^2, xz - w^2) \subset \mathbb{P}^2(k)$.

Solution:

1. Coordinate ring is $k[X] = k[x, x^{-1}]$, so the function field $k(X) = k(x)$ is of transcendence degree 1 and $\dim(X) = 1$.
2. Coordinate ring is $k[X] = k[x, y, z, w]/(xy - z^2, x + w - 1) \simeq k[x, y, z]/(xy - z^2) = k[\bar{x}, \bar{y}, \bar{z}]$, so the function field $k(X) = k(\bar{x}, \bar{y}, \bar{z})$ is of transcendence degree 2 (it is a quadratic extension of $k(\bar{x}, \bar{y})$) and so $\dim(X) = 2$.
3. Since $U = k^n$ is open in $X = \mathbb{P}^n(k)$ and $k(U) = k(X)$, we have $\dim(U) = \dim(X)$. Then $k(U) = \text{Frac}(k[x_1, \dots, x_n]) = k(x_1, \dots, x_n)$ so $\dim(U) = n$.

More concretely, one can say that the function field of $k(X) = \mathbb{P}^n(k)$ is (by definition) the field of fractions of the form $\frac{a}{b}$ where $a, b \in k[x_1, \dots, x_{n+1}]$ are homogeneous of same degree (it is a subfield of $\text{Frac}(k_h[X])$). We have an isomorphism by de/homogenizing

$$\begin{aligned} k(X) &\longrightarrow k(x_1, \dots, x_n), \\ \frac{a(x_1, \dots, x_{n+1})}{b(x_1, \dots, x_{n+1})} &\longmapsto \frac{a(x_1, \dots, x_n, 1)}{b(x_1, \dots, x_n, 1)}. \end{aligned}$$

The inverse is given by

$$\begin{aligned} k(x_1, \dots, x_n) &\longrightarrow k(X), \\ \frac{f(x_1, \dots, x_n)}{g(x_1, \dots, x_n)} &\longmapsto \frac{x_{n+1}^k f(x_1/x_{n+1}, \dots, x_n/x_{n+1})}{x_{n+1}^l g(x_1/x_{n+1}, \dots, x_n/x_{n+1})}, \end{aligned}$$

where the powers are such that both polynomials are homogeneous of the same degree.

4. The dimension is the same as the dimension of the intersection with the open set $X \cap U_x = V_a(y - w^2, z - w^2) \subset k^3$. The coordinate ring is $k[X \cap U_x] = k[y, z, w]/(y - w^2, z - w^2) \simeq k[y, w]/(y - w^2) = k[\bar{y}, \bar{w}]$. The function field is $k(\bar{y}, \bar{w})$, which is quadratic extension of $k(\bar{y})$. Thus $\dim(X) = 1$.

2. (a) Let $K \subseteq K'$ be two finitely generated field extension of k . Show that $\text{tr.deg}_k(K) \leq \text{tr.deg}_k(K')$.
- (b) Let $f: X \dashrightarrow Y$ be a dominant rational map between quasi-projective. Show that $\dim(Y) \leq \dim(X)$. Give an example where $\dim(Y) < \dim(X)$.
If it is a finite morphism, then $f(X)$ is dominant by definition. In that case, the condition that $k[X]$ is integral over $k[Y]$ implies that $k(X)$ is an algebraic extension of $k(Y)$, so $\dim(X) = \dim(Y)$.

Solution:

- (a) Suppose that $n = \text{tr.deg}_k(K)$. Then K has a transcendence basis $\{x_1, \dots, x_n\}$ such that K is algebraic over $k(x_1, \dots, x_n)$. The set $\{x_1, \dots, x_n\}$ is algebraically independent subset of K , and thus also of K' . This shows that $\text{tr.deg}_k(K) \leq \text{tr.deg}_k(K')$.
- (b) Let $f: U \rightarrow V$ be a morphism representing f , where U, V are open affine subsets such that $f(U) \subseteq V$. If $f: U \rightarrow V$ is dominant then $k[V]$ is isomorphic to the subring $f^*k[V] \subseteq k[U]$. For affine varieties, we have $k(U) = \text{Frac}(k[U])$, so it follows that $k(V)$ is isomorphic to a subfield of $k(U)$. From part a), it follows that $\dim(Y) = \dim(V) \leq \dim(U) = \dim(X)$.

An example where strict inequality occurs could be the projection $(x, y) \rightarrow y$ from $X = k^2$ to $Y = k$. It is surjective and $1 = \dim(Y) < \dim(X) = 2$.

3. (a) Let $f, g: X \rightarrow Y$ be morphisms between two quasi-projective varieties X and Y . If f and g agree on U , show that they agree on X .
- (b) Verify that the relation that we used to define a rational map is an equivalence relation.

Solution:

- (a) Let us first show the statement for affine varieties $X \subseteq k^n$ and $Y \subseteq k^m$. Let $Z := \{p \in X \mid f(p) = g(p)\}$ be the set where f and g agree. Let us show that this set is closed. The morphisms are represented by two polynomial maps $f = (f_1, \dots, f_m)$ and $g = (g_1, \dots, g_m)$ with $f_i, g_i \in k[x_1, \dots, x_n]$. Then $p \in Z$, if and only if $f_i(p) - g_i(p) = 0$ for $i \in \{1, \dots, m\}$. It follows that $Z = X \cap V(f_1 - g_1) \cap \dots \cap V(f_m - g_m)$ is closed in X . On the other hand, if f, g agree on U , then $U \subseteq Z$. But U is open so it is dense, and it follows that $X = \overline{U} \subseteq Z \subseteq X$. This shows that $Z = X$, and that f and g agree everywhere.

Let $(U_\alpha)_{\alpha \in A}$ be an affine cover of X and $(V_\alpha)_{\alpha \in A}$ a cover of Y , small enough that $f(U_\alpha) \subseteq V_\alpha$. If f and g agree on U , then $f|_{U_\alpha}$ and $g|_{U_\alpha}$ agree on each $U \cap U_\alpha$. For each $\alpha \in A$ we have now two morphisms $f|_{U_\alpha}, g|_{U_\alpha} : U_\alpha \rightarrow V_\alpha$ of affine varieties, that agree on the nonempty open set $U \cap U_\alpha \subseteq U_\alpha$. We deduce from the first part of the proof that $f|_{U_\alpha}$ and $g|_{U_\alpha}$ agree on each affine piece U_α , for all $\alpha \in A$. This shows that f, g agree everywhere on X .

- (b) For the equivalence relation, the only thing to check is the transitivity; the reflexivity and the symmetry are immediate. Let $f_i : U_i \rightarrow k$ be three morphisms on nonempty open subsets. If f_1 and f_2 agree on $U_1 \cap U_2$, and f_2 and f_3 agree on $U_2 \cap U_3$, then f_1 and f_3 agree on $U_1 \cap U_2 \cap U_3$. This set is dense in $U_1 \cap U_3$, so by part a) they agree on $U_1 \cap U_3$.

4. Let $\phi : \mathbb{P}^2(k) \dashrightarrow \mathbb{P}^3(k)$ be the rational map represented by

$$[x : y : z] \mapsto [xy : xz : y^2 : yz].$$

Determine the domain U where ϕ is defined. Show that $\phi : \mathbb{P}^2(k) \dashrightarrow X$ is birational, where $X = V(z_1z_4 - z_2z_3)$. Deduce that X is the closure of $\phi(U)$.

Solution: The function is well-defined, except if $xy = xz = y^2 = yz = 0$. This is the case if $x = y = 0$, or $y = z = 0$. So the domain of ϕ is $U = \mathbb{P}^2(k) \setminus \{[0 : 0 : 1], [1 : 0 : 0]\}$. The image of U lies in X , since

$$z_1z_4 - z_2z_3 = (xy)(yz) - xzy^2 = 0.$$

Let $U_y \subseteq U$ be the open set where $y \neq 0$ and $V := U_{z_3} \subseteq \mathbb{P}^3(k)$ where $z_3 \neq 0$. The morphism ϕ restricts to a morphism $\phi : U_y \rightarrow X \cap V$. Let us define an inverse morphism $\psi : X \cap V \rightarrow U_y$ by

$$\psi([z_1 : z_2 : z_3 : z_4]) = [z_1 : z_3 : z_4].$$

We have

$$\begin{aligned}
\phi(\psi([z_1 : z_2 : z_3 : z_4])) &= \phi([z_1 : z_3 : z_4]) \\
&= [z_1 z_3 : z_1 z_4 : z_3^2 : z_3 z_4] \\
&= [z_1 z_3 : z_2 z_3 : z_3^2 : z_3 z_4] \\
&= [z_1 : z_2 : z_3 : z_4],
\end{aligned}$$

where we used that $z_1 z_4 = z_2 z_3$ on X , and that we can divide by z_3 on $X \cap V$. Conversely, we have

$$\psi(\phi([x : y : z])) = \psi([xy : xz : y^2 : yz]) = [xy : y^2 : yz] = [x : y : z].$$

This shows that $\phi: U_y \rightarrow X \cap V$ is an isomorphism, so ϕ is birational.

In particular, this shows that $\overline{f(U_y)}$ is 2-dimensional variety contained in X , so that we conclude $X = \overline{f(U)}$. (We showed that $Y \subseteq X$ implies $\dim(Y) \leq \dim(X)$ with equality iff $X = Y$).

5. Let $X \subseteq k^2$ be the curve $X = V(f)$ where $y^2 - x^3(x-1)^2$ in k^2 .
- (a) Determine the points where X is smooth.
 - (b) Use a sequence of blow-ups to find a smooth curve Y in k^2 that is birational to Y . Write the birational map and determine its inverse.
 - (c) Deduce that X is a rational curve, and find a birational map from $X \dashrightarrow k$.

Solution:

- (a) A point $p \in X$ is singular if the derivatives of f

$$\begin{aligned}
\partial_x f &= -2x(x-1)(5x-3) = 0 \\
\partial_y f &= 2y = 0
\end{aligned}$$

both vanish. This gives the points $\{(0,0), (1,0), (3/5,0)\}$. The points $(3/5,0)$ is not in X , so the singular points are $\{(0,0), (1,0)\}$.

- (b) First, we resolve the singularity at $(0,0)$. Let $f(x,xy) = (xy)^2 - x^3(x-1)^2 = x^2(y^2 - x(x-1)^2)$. The blow-up at $(0,0)$ is the curve $X_1 = V(f_1)$ where

$$f_1(x,y) = y^2 - x(x-1)^2, \tag{1}$$

and the map $\varphi_1: X_1 \rightarrow X$ defined by $\varphi_1(x, y) = (x, xy)$ is birational. Looking at the derivatives of f_1 :

$$\begin{aligned}\partial_x f_1 &= -(x-1)(3x-1) = 0 \\ \partial_y f_2 &= 2y = 0.\end{aligned}$$

The curve Y_1 has still as singularity at $(1, 0)$, that can be resolved by the blowup at $(1, 0)$ by the map $(x, y) \mapsto (1+x, xy)$. We have

$$f_1(1+x, xy) = (xy)^2 - (1+x)x^2 = x^2(y^2 - 1 - x). \quad (2)$$

So the blowup of X_1 is the curve $X_2 = V(f_2)$ where $f_2(x, y) = y^2 - 1 - x$. This curve is smooth and the map $\varphi_2: X_2 \rightarrow X_1$ defined by $\varphi_2(x, y) = (1+x, xy)$ is birational.

Thus the curve $Y = X_2$ is smooth and the map

$$\begin{aligned}\varphi &:= \varphi_2 \circ \varphi_1: Y \rightarrow X \\ (x, y) &\mapsto (1+x, (1+x)xy)\end{aligned}$$

is birational. The inverse is defined on the open set $U := X \setminus \{(0, 0), (1, 0)\}$ by

$$\begin{aligned}\psi: U &\rightarrow Y \\ (u, v) &\mapsto \left(u-1, \frac{v}{u(u-1)}\right).\end{aligned}$$

- (c) The curve $Y = (y^2 - 1 - x)$ is a rational curve. The coordinate ring is $k[Y] = k[\bar{x}, \bar{y}]$, so the function field is $k(Y) = k(\bar{x}, \bar{y})$. Let us set $t := \frac{\bar{x}+1}{\bar{y}}$. Clearly $k(t) \subseteq k(Y)$, let us show the other inclusion. We have

$$t^2 = \frac{(\bar{x}+1)^2}{\bar{y}^2} = \frac{(\bar{x}+1)^2}{\bar{x}+1} = \bar{x}+1,$$

so that

$$\bar{x} = t^2 - 1, \quad \bar{y} = \frac{\bar{x}+1}{t} = t.$$

This shows that $k(Y) = k(t)$, and it induces a birational map

$$k \rightarrow Y \quad t \mapsto (t^2 - 1, t).$$

This is actually an isomorphism, the inverse is the projection $\pi_2(x, y) = y$. By composing with the previous birational morphism $\varphi: Y \rightarrow X$ we get a birational morphism

$$k \rightarrow X, \quad t \mapsto (t^2, t^3(t^2 - 1)).$$

The inverse is the rational map $X \dashrightarrow k$ represented by

$$\begin{aligned}\pi_2 \circ \psi: U &\rightarrow k \\ (u, v) &\mapsto \frac{v}{u(u-1)}.\end{aligned}$$